

# Nigeria results

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PRELIMINARY — PLEASE DO NOT CIRCULATE  
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## **Abstract**

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# 1 Introduction

## Tables

### Balance and Attrition

Table 1: Nigeria Baseline Balance

| Variable   | Treat mean | Control mean | Diff   | SE       | p-value           |
|------------|------------|--------------|--------|----------|-------------------|
| bl_score   | 80.171     | 82.598       | -2.985 | ( 2.570) | .2520921542422807 |
| has_t2_mte | 0.622      | 0.634        | -0.026 | ( 0.041) | .5255423366734244 |
| has_t2_ete | 0.691      | 0.677        | -0.001 | ( 0.034) | .978987091575142  |
| has_t3_mte | 0.470      | 0.516        | -0.103 | ( 0.073) | .1639883600625275 |
| has_t3_ete | 0.650      | 0.511        | 0.135  | ( 0.084) | .1168849892737642 |

Notes: Follow-up indicators are measured from baseline sample. Diff column uses OLS with constituency and grade FE, clustered by academy.

Table 2: Nigeria Attrition from Baseline by Term and Window

| Wave/Outcome | Treat retain | Control retain | Diff   | SE       | p-value           |
|--------------|--------------|----------------|--------|----------|-------------------|
| t2_mte_math  | 0.788        | 0.800          | -0.029 | ( 0.050) | .5692328141075447 |
| t2_mte_num   | 0.790        | 0.753          | 0.025  | ( 0.080) | .7591735038805077 |
| t2_ete_math  | 0.873        | 0.855          | -0.017 | ( 0.031) | .5860965324065082 |
| t2_ete_num   | 0.879        | 0.854          | -0.010 | ( 0.031) | .7419433312069506 |
| t3_mte_math  | 0.589        | 0.636          | -0.145 | ( 0.095) | .135659466228529  |
| t3_mte_num   | 0.594        | 0.627          | -0.130 | ( 0.093) | .1695486919117143 |
| t3_ete_math  | 0.812        | 0.619          | 0.170  | ( 0.104) | .1106300306561516 |
| t3_ete_num   | 0.797        | 0.597          | 0.183  | ( 0.112) | .1099210629023511 |

Notes: Retention indicators are defined among students with non-missing baseline score.

### Main Effects by Wave

### Over-Time Specifications

### IRT Outcomes

### Interactions by Group Color

Table 3: Effect on T2 Midterm Outcomes

|              | (1)               | (2)                | (3)                   |
|--------------|-------------------|--------------------|-----------------------|
|              | Math              | Numeracy           | Math + Numeracy Index |
| Treatment    | 0.054<br>( 0.100) | -0.078<br>( 0.147) | 0.011<br>( 0.108)     |
| Control Mean | 0.000             | -0.000             | -0.017                |
| N            | 1868              | 1809               | 1781                  |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. \*\*\*, \*\*, and \* indicate 1%, 5%, and 10%.

Table 4: Effect on T2 Endterm Outcomes

|              | (1)               | (2)               | (3)                   |
|--------------|-------------------|-------------------|-----------------------|
|              | Math              | Numeracy          | Math + Numeracy Index |
| Treatment    | 0.151<br>( 0.115) | 0.135<br>( 0.083) | 0.140*<br>( 0.082)    |
| Control Mean | -0.000            | 0.001             | 0.000                 |
| N            | 2030              | 2035              | 2029                  |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. \*\*\*, \*\*, and \* indicate 1%, 5%, and 10%.

Table 5: Effect on T3 Midterm Outcomes

|              | (1)               | (2)               | (3)                   |
|--------------|-------------------|-------------------|-----------------------|
|              | Math              | Numeracy          | Math + Numeracy Index |
| Treatment    | 0.045<br>( 0.115) | 0.096<br>( 0.110) | 0.071<br>( 0.108)     |
| Control Mean | -0.000            | 0.000             | -0.004                |
| N            | 1446              | 1440              | 1424                  |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. \*\*\*, \*\*, and \* indicate 1%, 5%, and 10%.

Table 6: Effect on T3 Endterm Outcomes

|              | (1)               | (2)               | (3)                   |
|--------------|-------------------|-------------------|-----------------------|
|              | Math              | Numeracy          | Math + Numeracy Index |
| Treatment    | 0.039<br>( 0.150) | 0.104<br>( 0.144) | 0.061<br>( 0.117)     |
| Control Mean | -0.000            | 0.000             | 0.016                 |
| N            | 1660              | 1615              | 1615                  |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. \*\*\*, \*\*, and \* indicate 1%, 5%, and 10%.

Table 7: Effects Using Data Over Terms

|              | (1)               | (2)               | (3)                   |
|--------------|-------------------|-------------------|-----------------------|
|              | Math              | Numeracy          | Math + Numeracy Index |
| Treatment    | 0.158<br>( 0.119) | 0.016<br>( 0.119) | 0.088<br>( 0.102)     |
| Control Mean | 0.014             | 0.070             | 0.019                 |
| N            | 934               | 868               | 845                   |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. \*\*\*, \*\*, and \* indicate 1%, 5%, and 10%.

Table 8: Indices Over Time: T2 Midterm + T2 Endterm

|              | (1)               | (2)               | (3)                   |
|--------------|-------------------|-------------------|-----------------------|
|              | Math              | Numeracy          | Math + Numeracy Index |
| Treatment    | 0.119<br>( 0.088) | 0.020<br>( 0.095) | 0.090<br>( 0.079)     |
| Control Mean | -0.002            | -0.021            | -0.026                |
| N            | 1786              | 1729              | 1700                  |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. \*\*\*, \*\*, and \* indicate 1%, 5%, and 10%.

Table 9: Indices Over Time: T3 Midterm + T3 Endterm

|              | (1)               | (2)               | (3)                   |
|--------------|-------------------|-------------------|-----------------------|
|              | Math              | Numeracy          | Math + Numeracy Index |
| Treatment    | 0.062<br>( 0.137) | 0.022<br>( 0.090) | 0.023<br>( 0.092)     |
| Control Mean | 0.011             | 0.072             | 0.055                 |
| N            | 1121              | 1079              | 1063                  |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. \*\*\*, \*\*, and \* indicate 1%, 5%, and 10%.

Table 10: IRT-Based Outcomes (T3 ETE where item data available)

|              | (1)               | (2)               | (3)                   |
|--------------|-------------------|-------------------|-----------------------|
|              | Math              | Numeracy          | Math + Numeracy Index |
| Treatment    | 0.104<br>( 0.144) | 0.103<br>( 0.136) | 0.113<br>( 0.149)     |
| Control Mean | 0.000             | 0.176             | 0.220                 |
| N            | 1615              | 1737              | 1737                  |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. \*\*\*, \*\*, and \* indicate 1%, 5%, and 10%.

Table 11: T2 Midterm Effects by Group Color (Red/Blue/Yellow)

|                     | Math                  | Numeracy          | Math + Numeracy Index |
|---------------------|-----------------------|-------------------|-----------------------|
| Red group effect    | -0.747***<br>( 0.193) | 0.789<br>( 0.522) | 0.002<br>( 0.200)     |
| Blue group effect   | -0.747***<br>( 0.193) | 0.789<br>( 0.522) | 0.002<br>( 0.200)     |
| Yellow group effect | -0.747***<br>( 0.193) | 0.789<br>( 0.522) | 0.002<br>( 0.200)     |
| N                   | 745                   | 741               | 728                   |

Notes: Effects are from treatment-by-group interaction regressions; Red is base category. Controls and SE clustering as in baseline tables.

Table 12: T2 Endterm Effects by Group Color (Red/Blue/Yellow)

|                     | Math                 | Numeracy             | Math + Numeracy Index |
|---------------------|----------------------|----------------------|-----------------------|
| Red group effect    | 0.213<br>( 0.126)    | -0.104<br>( 0.129)   | 0.060<br>( 0.107)     |
| Blue group effect   | 0.213<br>( 0.126)    | -0.104<br>( 0.129)   | 0.060<br>( 0.107)     |
| Yellow group effect | 0.386***<br>( 0.093) | 1.262***<br>( 0.193) | 0.815***<br>( 0.121)  |
| N                   | 833                  | 839                  | 833                   |

Notes: Effects are from treatment-by-group interaction regressions; Red is base category. Controls and SE clustering as in baseline tables.

Table 13: T3 Midterm Effects by Group Color (Red/Blue/Yellow)

|                     | Math                 | Numeracy             | Math + Numeracy Index |
|---------------------|----------------------|----------------------|-----------------------|
| Red group effect    | 2.382***<br>( 0.145) | 1.432***<br>( 0.156) | 1.882***<br>( 0.138)  |
| Blue group effect   | 2.382***<br>( 0.145) | 1.432***<br>( 0.156) | 1.882***<br>( 0.138)  |
| Yellow group effect | -0.321<br>( 0.259)   | 0.270<br>( 0.199)    | -0.019<br>( 0.197)    |
| N                   | 583                  | 588                  | 577                   |

Notes: Effects are from treatment-by-group interaction regressions; Red is base category. Controls and SE clustering as in baseline tables.

Table 14: T3 Endterm Effects by Group Color (Red/Blue/Yellow)

|                     | Math                 | Numeracy            | Math + Numeracy Index |
|---------------------|----------------------|---------------------|-----------------------|
| Red group effect    | 0.279*<br>( 0.151)   | 0.883**<br>( 0.322) | 0.585***<br>( 0.200)  |
| Blue group effect   | 0.279*<br>( 0.151)   | 0.883**<br>( 0.322) | 0.585***<br>( 0.200)  |
| Yellow group effect | -0.786**<br>( 0.359) | -0.444<br>( 0.262)  | -0.599*<br>( 0.290)   |
| N                   | 810                  | 797                 | 797                   |

Notes: Effects are from treatment-by-group interaction regressions; Red is base category. Controls and SE clustering as in baseline tables.

Table 15: Over-Term Effects by Group Color (Red/Blue/Yellow)

|                     | Math                 | Numeracy             | Math + Numeracy Index |
|---------------------|----------------------|----------------------|-----------------------|
| Red group effect    | 0.519***<br>( 0.126) | 0.942***<br>( 0.115) | 0.704***<br>( 0.088)  |
| Blue group effect   | 0.519***<br>( 0.126) | 0.942***<br>( 0.115) | 0.704***<br>( 0.088)  |
| Yellow group effect | 0.519***<br>( 0.126) | 0.942***<br>( 0.115) | 0.704***<br>( 0.088)  |
| N                   | 437                  | 433                  | 416                   |

Notes: Effects are from treatment-by-group interaction regressions; Red is base category. Controls and SE clustering as in baseline tables.

Table 16: T2 Over-Time Indices by Group Color (Red/Blue/Yellow)

|                     | Math                  | Numeracy          | Math + Numeracy Index |
|---------------------|-----------------------|-------------------|-----------------------|
| Red group effect    | -0.267***<br>( 0.076) | 0.318<br>( 0.292) | 0.007<br>( 0.140)     |
| Blue group effect   | -0.267***<br>( 0.076) | 0.318<br>( 0.292) | 0.007<br>( 0.140)     |
| Yellow group effect | -0.267***<br>( 0.076) | 0.318<br>( 0.292) | 0.007<br>( 0.140)     |
| N                   | 718                   | 715               | 701                   |

Notes: Effects are from treatment-by-group interaction regressions; Red is base category. Controls and SE clustering as in baseline tables.

Table 17: T3 Over-Time Indices by Group Color (Red/Blue/Yellow)

|                     | Math                 | Numeracy             | Math + Numeracy Index |
|---------------------|----------------------|----------------------|-----------------------|
| Red group effect    | 1.359***<br>( 0.180) | 1.235***<br>( 0.158) | 1.258***<br>( 0.127)  |
| Blue group effect   | 1.359***<br>( 0.180) | 1.235***<br>( 0.158) | 1.258***<br>( 0.127)  |
| Yellow group effect | -0.465<br>( 0.323)   | -0.008<br>( 0.161)   | -0.209<br>( 0.221)    |
| N                   | 540                  | 533                  | 522                   |

Notes: Effects are from treatment-by-group interaction regressions; Red is base category. Controls and SE clustering as in baseline tables.



Table 18: IRT Outcome Effects by Group Color (Red/Blue/Yellow)

|                     | Math                | Numeracy             | Math + Numeracy Index |
|---------------------|---------------------|----------------------|-----------------------|
| Red group effect    | 0.883**<br>( 0.322) | 0.651***<br>( 0.109) | 0.714***<br>( 0.119)  |
| Blue group effect   | 0.883**<br>( 0.322) | 0.651***<br>( 0.109) | 0.714***<br>( 0.119)  |
| Yellow group effect | -0.444<br>( 0.262)  | -0.075<br>( 0.237)   | -0.082<br>( 0.260)    |
| N                   | 797                 | 733                  | 733                   |

Notes: Effects are from treatment-by-group interaction regressions; Red is base category. Controls and SE clustering as in baseline tables.